Shivam Chopra

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Summary

Mechanical Engineer and Technical Lead with a PhD in Robotics, specializing in system-level mechanical architecture and electromechanical integration for wearable and robotic systems. Leads early-stage feasibility and NPI programs from subsystem architecture through preclinical readiness by translating requirements and risk into executable build, validation, and test strategies. Experience in driving cross-functional integration across mechanical, firmware, electronics, and manufacturing teams.

Experience

Dexcom — Senior Medical Device Engineer — Dec 2022 – Present

Subsystem Technical Lead - Next Generation Wearable Platform — Apr 2025 – Present

- Owned system-level development of a critical electro-mechanical sensor-connect subsystem for a new wearable electronics architecture, spanning process design, build execution, validation strategy, and cross-functional integration.
- Delivered ∼2,000 subsystem assemblies enabling system integration and validation campaigns by establishing end-to-end build readiness including fixtures, work instructions, inspection criteria, and technician training.
- Stabilized and transferred a sensitive manufacturing process to an external partner, removing a major feasibility bottleneck and enabling scalable subsystem builds while maintaining IP boundaries and quality expectations.
- Led system-level risk-based validation strategy by translating requirements and failure modes into targeted test methods and decision-ready evidence for program stakeholders.
- Enabled end-to-end connectivity demonstrations by coordinating production firmware integration under cybersecurity and data-protection constraints across hardware and software teams.
- Acted as the on-site technical point of accountability at a partner location, aligning design, manufacturing, testing, and data workflows to drive execution and resolve issues quickly.

R&D Technology Evaluation, Advanced Technology — Dec 2022 – Mar 2025

- Designed and delivered electromechanical test platforms and prototype systems supporting wearable sensor R&D, integrating precision motion, actuation, and measurement to enable repeatable system characterization from feasibility through DVT.
- Led cross-functional validation efforts across bench, preclinical, and clinical environments for a wearable patch program enabling extension of product wear duration from 10 to 15 days.
- Drove mechanical design release quality by leading GD&T and tolerance analysis for multi-part assemblies; produced detailed drawings, defined fits/clearances, and partnered with suppliers on manufacturability and process capability (DFM/DFA).
- Developed bench and preclinical test methods to characterize wear, sensor motion, and stability; executed structured mechanical root-cause investigations and communicated findings to cross-functional stakeholders.
- Accelerated issue resolution by rapidly developing new test methods for failure investigations (including accelerated fatigue characterization on short timelines) and using data-driven experiments to isolate root causes.

Adsys Controls Inc. — Senior Robotics Engineer — Apr 2022 – Nov 2022

- Designed and prototyped electromechanical assemblies for outdoor consumer products, owning enclosure architecture and rapid design iteration using CAD and additive manufacturing.
- Led testing and performance validation of a precision fast steering mirror system, evaluating stability, repeatability, and alignment through benchtop experimentation.

Gravish Lab, UC San Diego — PhD Researcher — Underactuated Robots for Granular Media — Jul 2018 – Mar 2022

- Owned end-to-end development of an underactuated robotic system operating in granular and underwater environments, from mechanism architecture through fabrication, integration, and experimental validation.
- Built constrained-volume, untethered electromechanical prototypes integrating actuation and sensing; balanced stiffness, compliance, and robustness to achieve reliable locomotion under granular material load.
- Designed and executed tests using custom experimental setups integrating high-speed imaging, locomotion control and force sensing to characterize performance limits, failure modes, and robot-environment interactions.
- Developed automated test rigs and data-analysis workflows to accelerate hypothesis-to-result cycles, improve

experimental repeatability, and support data-driven iteration across multiple prototype generations.

Gravish Lab, UC San Diego — Masters Research — Flapping Wing Actuator Design Dec 2016 – Jun 2018

- Designed and fabricated penny-scale flapping mechanisms with piezoelectric actuation; engineered lightweight frames out of lamination techniques, flexures, and transmissions for high-frequency motion and durability.
- Integrated custom on-board sensing and closed-loop control concepts to study actuator response under environmental perturbations

Skills

- **Mechanical Design & Architecture:** System-level mechanical architecture, tolerance analysis, load paths, material selection, GD&T (ASME Y14.5), DFM/DFA
- **Verification, Reliability & Debug:** Mechanical validation and life testing, wear characterization, custom test platforms, root-cause analysis
- **Robotics & Automation:** Robotic mechanism and kinematic design, actuator and sensor integration, compliant/adaptive mechanisms, constrained-volume robotic assemblies, automated test platforms, haptics, soft robotics
- **CAD, Analysis & Engineering Tools:** SolidWorks, Onshape; PDM best practices; FEA-informed sizing and risk decisions (Ansys); DOE and variability analysis (JMP)
- **Machine Vision & Imaging:** Cognex In-Sight Explorer; machine vision camera and illumination strategies, image processing for feature isolation in in-line process tools and standalone systems
- **Prototyping & Fabrication:** Machining, 3D printing, laser cutting, molding, laminate fabrication, soft-material prototyping
- **Programming:** LabVIEW, MATLAB, Python, C#, JMP, Arduino

Education

UC San Diego, CA — PhD in Mechanical Engineering (Robotics) Mar 2022

- microMBA (Business), Jun 2021
- MS Mechanical Engineering, Jun 2018

Punjab Engineering College, India — BE Mechanical Engineering Aug 2016

- Chancellor's Gold Medal Recipient

Selected Publications

For full list, visit Google Scholar.

- Chopra S., Gravish N. (2019), Piezoelectric actuators with on-board sensing for micro-robotic applications, Smart Materials and Structures, IOP.
- Chopra S., Tolley M. T., Gravish N. (2020), Granular Jamming Feet Enable Improved Foot-Ground Interactions for Robot Mobility, IEEE Robotics and Automation Letters.
- Chopra S., et al. (2023), Toward Robotic Sensing and Swimming in Granular Environments using Underactuated Appendages, Advanced Intelligent Systems.

Press Coverage

- Bot inspired by baby turtles can swim under the sand - UCSD Today
- Sand-swimming robot inspired by baby sea turtles - New Atlas
- Turtle-like robot swims under sand - IoT World Today
- Turtle-inspired robot detects obstacles in sand - Interesting Engineering
- Robot digs through sand like a turtle - The Robot Report
- New robot inspired by baby sea turtles - Tech Times
- Robot swims under sand like sea turtle hatchlings - Earth.com
- New robot explores secrets of sand - Knowridge
- Bot inspired by baby turtles - Life Technology